Develop a LaTeX script to present an algorithm in the document using algorithm/algorithmic/algorithm2e library.

Code:

```
\documentclass{article}
\usepackage{algorithm}
\usepackage{algpseudocode}
\begin{document}
\begin{algorithm}
\caption{Bubble Sort}
\begin{algorithmic}[1]
\Procedure{BubbleSort}{$A, n$}
\For{$i \gets 0$ to $n-1$}
\For{$j \gets 0$ to $n-1-i$}
\If{$A[j] > A[j+1]$}
\State Swap $A[j]$ and $A[j+1]$
\EndIf
\EndFor
\EndFor
\EndProcedure
\end{algorithmic}
\end{algorithm}
\end{document}
```

Output:

Algorithm 1 Bubble Sort

procedure BUBBLESORT(A, n)
    for $i \leftarrow 0$ to $n - 1$ **do**
        for $j \leftarrow 0$ to $n - 1 - i$ **do**
            if $A[j] > A[j + 1]$ **then**
                Swap $A[j]$ and $A[j + 1]$
            end if
        end for
    end for
end procedure

Experiment 12:

Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

Code:

```
\documentclass[6pt,a4paper]{report}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage{graphicx}
\usepackage[left=3cm,right=3cm,top=2cm,bottom=2cm]{geometry}
\author{Dr.Sajithra Varun}
\title{Artificial Intelligence}
\begin{document}

\maketitle
\chapter{Introduction}
\section*{What is AI?}
"\textbf{AI}"  refers to the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, and decision-making.

AI is a broad field that encompasses a wide range of techniques and technologies aimed at enabling machines to mimic human-like intelligence.

AI is the backbone of innovation in modern computing, unlocking value for individuals and businesses. For example, optical character recognition (OCR) uses AI to extract text and data from images and documents, turns unstructured content into business-ready structured data, and unlocks valuable insights.


\section*{Key Concepts}
\begin{itemize}
\item\textbf{Learning}:AI systems can learn from data and improve their performance over time.
\item\textbf{Reasoning:}
AI systems can draw conclusions and make inferences based on available information.
\item\textbf{Problem-solving:}
AI systems can identify and solve problems, often by using algorithms and data analysis.
\item\textbf{Decision-making:}
AI systems can make decisions based on available information and learned patterns.
\end{itemize}
\section*{Examples of AI Applications:}
\begin{itemize}
\item Voice assistants: Siri, Alexa, and Google Assistant rely on AI to understand and respond to voice commands.
```

\item Recommendation systems: Netflix, Amazon, and Spotify use AI to suggest content or products based on user preferences.
\item Self-driving cars: Autonomous vehicles use AI to navigate roads, detect obstacles, and make driving decisions.
\item Chatbots: AI-powered chatbots are used to provide customer support and answer questions.
\item Image recognition: AI can be used to identify objects, faces, and scenes in images.

\end{itemize}
\section*{Types of AI:}

\begin{itemize}
\item Narrow AI (Weak AI): Designed to perform a specific task, like a chatbot or a recommendation system.\\
\item General AI (Strong AI): A hypothetical type of AI that can perform any intellectual task that a human being can.\\
\item Superintelligence: A hypothetical AI that is far more intelligent than humans. \\
\end{itemize}

\chapter{Goals}

\begin{small}
\section*{AI and Machine Learning}

\subsection*{Reasoning and problem-solving}
Early researchers developed algorithms that imitated step-by-step reasoning that humans use when they solve puzzles or make logical deductions.By the late 1980s and 1990s, methods were developed for dealing with uncertain or incomplete information, employing concepts from probability and economics.\\

Many of these algorithms are insufficient for solving large reasoning problems because they experience a "combinatorial explosion": They become exponentially slower as the problems grow. Even humans rarely use the step-by-step deduction that early AI research could model. They solve most of their problems using fast, intuitive judgments. Accurate and efficient reasoning is an unsolved problem. \\

\subsection*{Knowledge representation}

Knowledge representation and knowledge engineering allow AI programs to answer questions intelligently and make deductions about real-world facts. Formal knowledge representations are used in content-based indexing and retrieval,scene interpretation, clinical decision support, knowledge discovery (mining "interesting" and actionable inferences from large databases), and other areas.\\

A knowledge base is a body of knowledge represented in a form that can be used by a program. An ontology is the set of objects, relations, concepts, and properties used by a particular domain of knowledge.Knowledge bases need to represent things such as objects, properties, categories, and relations between objects; situations, events, states, and time;causes and effects;knowledge about knowledge (what we know about what other people know); default reasoning (things that humans assume are true until they are told differently and will remain true even when other facts are changing); and many other aspects and domains of knowledge.

\subsection*{Planning and decision-making}
An "agent" is anything that perceives and takes actions in the world. A rational agent has goals or preferences and takes actions to make them happen. In automated planning, the agent has a specific goal. In automated decision-making, the agent has preferences—there are some situations it would prefer to be in, and some situations it is trying to avoid. The decision-making agent assigns a number to each situation (called the "utility") that measures how much the agent prefers it. For each possible action, it can calculate the "expected utility": the utility of all possible outcomes of the action, weighted by the probability that the outcome will occur. It can then choose the action with the maximum expected utility.\\

In classical planning, the agent knows exactly what the effect of any action will be.In most real-world problems, however, the agent may not be certain about the situation they are in (it is "unknown" or "unobservable") and it may not know for certain what will happen after each possible action (it is not "deterministic"). It must choose an action by making a probabilistic guess and then reassess the situation to see if the action worked.\\

\subsection*{Learning}
Machine learning is the study of programs that can improve their performance on a given task automatically. It has been a part of AI from the beginning.\\

There are several kinds of machine learning. Unsupervised learning analyzes a stream of data and finds patterns and makes predictions without any other guidance. Supervised learning requires labeling the training data with the expected answers, and comes in two main varieties: classification (where the program must learn to predict what category the input belongs in) and regression (where the program must deduce a numeric function based on numeric input).\\

\subsection*{Natural Language Processing}

Natural language processing (NLP) allows programs to read, write and communicate in human languages such as English. Specific problems include speech recognition, speech synthesis, machine translation, information extraction, information retrieval and question answering.\\

Early work, based on Noam Chomsky's generative grammar and semantic networks, had difficulty with word-sense disambiguation unless restricted to small domains called "micro-worlds" (due to the common sense knowledge problem). Margaret Masterman believed that it was meaning and not grammar that was the key to understanding languages, and that thesauri and not dictionaries should be the basis of computational language structure.\\
\chapter{Results}

This is the results section.
\chapter{Discussion}
This is the discussion section.
\chapter{Conclusion}
This is the conclusion.
\end{small}

\end{document}

Output:

Artificial Intelligence

Dr.Sajithra Varun

March 25, 2025

Chapter 1

Introduction

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Chapter 2

Goals

AI and Machine Learning

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Chapter 3

Results

This is the results section.

Chapter 4

Discussion

This is the discussion section.

Chapter 5

Conclusion

This is the conclusion.